ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – Case Study Based Assignment

DWDM Metro Link Commissioning: From Field Measurements to a Standards-Based Upgrade Decision

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Course Outcome	CO5
Nature of Assessment	Case Study Based Assignment
Maximum Marks	100
Bloom's Levels	K4 – Analyze | K5 – Evaluate | K6 – Create/Design
Knowledge Profile	WK1, WK3, WK5

Case Study Summary

A metro operator's 60 km G.652.D link carrying eight 10 Gb/s NRZ channels on a 100 GHz ITU-T G.694.1 grid, amplified only by a booster EDFA, shows Channel 3 performing to expectation while Channel 5 shows a partially closed eye and high BER; power-meter data suggest even Channel 3's margin is below the required 3 dB. OTDR, OSA, power-meter and eye-diagram data are available, and the operator wants to upgrade the same fibre pair to 1.6 Tb/s, spectrally and energy efficiently, without re-laying fibre. This report follows REQUIREMENT → MEASUREMENT → ANALYSIS → ROOT-CAUSE IDENTIFICATION → REPAIR AND UPGRADE → VERIFICATION → DEPLOYMENT DECISION.

Given Data

Parameter	Value
Fibre	60 km ITU-T G.652.D, group index n=1.4682, 0.21 dB/km @1550 nm
Channel plan / launch power	8 × 10 Gb/s NRZ, 100 GHz grid, +1 dBm/channel at line output
Demultiplexer loss	6.0 dB
Design allowances	4 connectors × 0.5 dB; 13 splices × 0.05 dB
Receiver sensitivity	–24 dBm @ BER 10■¹² (valid for ER ≥ 8.2 dB); margin ≥ 3 dB required
Dispersion / PMD	D = 17 ps/(nm·km); source rms width 0.03 nm; PMD coeff. 0.1 ps/√km
Booster EDFA	NF = 5 dB; –6 dBm/channel at input; required OSNR 20 dB (0.1 nm)
System A (Ch 3)	Q = 7.6; ER = 9.0 dB; DJ = 0.18 UI; RJ = 0.012 UI; eye open
System B (Ch 5)	Q = 4.7; ER = 4.5 dB; DJ = 0.30 UI; RJ = 0.022 UI; eye partially closed

STAGE 1 – Performance Requirement Statement (10 Marks)

1.1 Measurement Instruments and What Each Validates

Instrument	Validates on this link
Optical power meter	Absolute launch/received power, as-built connector/splice loss (bypasses OTDR dead-zone limits)
Optical Spectrum Analyzer (OSA)	Per-channel OSNR, channel wavelength accuracy on the G.694.1 grid, amplifier gain flatness
OTDR	Fault location, event loss/reflectance, fibre attenuation uniformity along the 60 km span
Eye-diagram / BER tester	Q-factor, extinction ratio, jitter (DJ/RJ), eye opening, direct BER measurement
Dispersion / PMD analyzer	Chromatic-dispersion-induced penalty and PMD-induced pulse spreading vs bit-period tolerance

1.2 Budget Components to Include

- **Loss budget:** fibre attenuation, connector loss, splice loss, demultiplexer insertion loss.
- **Penalty budget:** chromatic-dispersion penalty (pulse broadening vs bit period) and extinction-ratio penalty (reduced effective signal swing).
- **Noise budget:** booster-EDFA ASE noise, expressed as received OSNR, compared against the 20 dB requirement.

1.3 Interpreting Eye/Jitter/ER Symptoms

A partially closed eye with high jitter and low extinction ratio (Channel 5: Q=4.7, ER=4.5 dB) indicates both reduced vertical (amplitude/noise) margin and reduced horizontal (timing) margin, which will push the measured BER far above the 10■¹² target (quantified in Stage 4), whereas Channel 3's higher Q (7.6) and ER (9.0 dB) indicate a healthy eye likely to meet BER even though its power margin alone may not.

1.4 OTDR Dynamic Range, Event Dead Zone (EDZ), Attenuation Dead Zone (ADZ)

- **Dynamic range (DR):** the maximum one-way loss the OTDR can characterise with adequate SNR; must exceed the span's one-way trace loss plus a safety margin (here, trace loss + 6 dB) to reliably see the far end.
- **Event dead zone (EDZ):** the minimum separation at which a second *reflective* event can be distinguished (detected) from a first.
- **Attenuation dead zone (ADZ):** the longer minimum separation needed to accurately *measure the loss* of a second event after a first — larger than EDZ because the detector must recover from saturation after a reflective peak.

1.5 Standards and Splice/Connector Choices

ITU-T G.652.D defines the standard single-mode fibre's attenuation/dispersion characteristics assumed here; ITU-T G.694.1 defines the DWDM frequency grid (193.1 THz anchor, 100/50/25/12.5 GHz spacings) used both for the existing 8-channel plan and the upgrade options. Fusion splicing gives much lower loss and negligible reflectance than mechanical splicing, and connector choice (SC/LC/MPO/CS, UPC vs APC) affects insertion loss and back-reflection — all of which bound the repair and upgrade strategy.

1.6 Practical Constraints

- No fibre re-laying — all gains must come from repairing/upgrading existing plant and equipment.
- ≥3 dB system margin required at −24 dBm receiver sensitivity.
- The 1.6 Tb/s upgrade must be spectrally efficient (fit within the existing C-band/fibre pair) and energy-efficient (minimise pJ/bit).

Stage 1 Deliverable – Performance Requirement Statement

The repaired and upgraded 60 km link must first achieve, on the legacy 8×10 Gb/s system, an as-built loss not exceeding the 21.25 dB design budget, a system margin ≥3 dB at −24 dBm sensitivity, OSNR ≥20 dB, PMD within 10% of the bit period, and BER ≤10■¹² (Q≥7.03) on every channel; and must then support a 1.6 Tb/s upgrade on the same fibre pair, selected by G.694.1-compliant spectral efficiency, adequate OSNR margin on the repaired link, and minimum energy per bit — all without re-laying fibre.

STAGE 2 – Link-Budget, Dispersion and OSNR Configuration (20 Marks)

2.1 Fibre Attenuation and As-Built Total Link Loss

Fibre attenuation = 60 km × 0.21 dB/km = ***12.6 dB***

Design total loss = 12.6 + (4×0.5) + (13×0.05) + 6.0 = 12.6 + 2.0 + 0.65 + 6.0 = ***21.25 dB***

Replacing the design allowances with the measured field values — connectors: event 3 (1.20 dB) + three 0.5 dB connectors including event 4 = 1.20 + 1.50 = 2.70 dB; splices: twelve 0.05 dB fusion splices including event 1 + event 2 mechanical splice (0.62 dB) = 0.60 + 0.62 = 1.22 dB —

As-built total loss = 12.6 + 2.70 + 1.22 + 6.0 = ***22.52 dB***

Excess over design = 22.52 – 21.25 = ***1.27 dB*** *(event 3 excess 0.70 dB + event 2 excess 0.57 dB = 1.27 dB — fully accounted for)*

2.2 Received Power, Dispersion Penalty and ER Penalty

*Received power (as-built) = +1 dBm – 22.52 dB = –****21.52 dBm*** *(common to both channels, shared fibre path)*

Dispersion penalty (shared, same fibre/bit-rate/source width for both channels): $\delta = 5\cdot\log[1+2\pi(B\cdot D\cdot L\cdot\sigma\lambda)^2]$, with $B\cdot D\cdot L\cdot\sigma\lambda = (10\times10^{-6})(17\times60\times0.03\times10^{-12}) = 0.306$:

$\delta = 5\cdot\log[1 + 2\pi(0.306)^2]$ = ***1.00 dB***

Extinction-ratio penalty $\delta_{ER} = 10\cdot\log[(ER+1)/(ER-1)]$ (ER in linear units):

Channel 3 (ER=9.0 dB → 7.94 linear): δ_{ER3} = ***1.10 dB***

Channel 5 (ER=4.5 dB → 2.82 linear): δ_{ER5} = ***3.22 dB***

Channel 5 carries a 2.12 dB larger ER penalty than Channel 3 — this is channel-specific since both channels share the same fibre and dispersion penalty.

2.3 System Margin vs the 3 dB Requirement

Quantity	Channel 3	Channel 5
Received power (as-built)	–21.52 dBm	–21.52 dBm
Dispersion penalty	1.00 dB	1.00 dB
ER penalty	1.10 dB	3.22 dB
Effective received power	–23.62 dBm	–25.75 dBm
Receiver sensitivity	–24 dBm	–24 dBm
System margin	0.38 dB	–1.75 dB
Meets ≥3 dB requirement?	No (short by 2.62 dB)	No (short by 4.75 dB, and negative)

Both channels fail the 3 dB margin requirement as-built — confirming the case-study observation that even Channel 3's margin is below spec, while Channel 5 is far worse, with a negative margin consistent with its partially closed eye and high BER (quantified in Stage 4).

2.4 OSNR and PMD Check

OSNR = 58 + P_in – NF = 58 + (–6) – 5 = ***47 dB*** *(≫ 20 dB requirement)*

Since the calculated OSNR (47 dB) exceeds the 20 dB requirement by a wide margin, **noise is not the limiting factor** on this link — the system is **power-and-penalty limited**, dominated by the 1.27 dB of excess fibre-plant loss and, for Channel 5, the additional 2.12 dB ER penalty.

PMD = 0.1 ps/√km × √60 km = ***0.77 ps****; 10% of 10 Gb/s bit period (100 ps) =* ***10 ps***

At 0.77 ps, PMD is less than 8% of the 10 ps tolerance — PMD is negligible and not a contributor to either channel's impairment.

Design Decision 1 – Link-Budget Configuration

I recommend treating this **as a loss-and-penalty-limited link, not a noise-limited one**, and correcting **excess fibre loss first (event 2 and event 3), followed by Channel 5's transmitter ER**, because the calculated OSNR (47 dB) has over 27 dB of margin above the 20 dB requirement while both channels fail the 3 dB power-margin target (Channel 3 by 2.62 dB, Channel 5 by 4.75 dB), and because the 1.27 dB fibre-plant excess loss is fully and exactly explained by events 2 and 3, with Channel 5's additional 2.12 dB ER penalty clearly isolated as a separate, channel-specific contribution.

STAGE 3 – OTDR, Splice and Connector Configuration (20 Marks)

3.1 Required Dynamic Range and Pulse-Width Selection

One-way trace loss (fibre+splices+connectors, excl. demux) = 12.6+2.70+1.22 = 16.52 dB

Required OTDR dynamic range = trace loss + 6 dB = ***22.52 dB***

Pulse Width	Dynamic Range	Event Dead Zone	Attenuation Dead Zone	Adequate for this link?
10 ns	20 dB	1 m	4 m	No — DR < 22.52 dB required
100 ns	27 dB	5 m	15 m	Yes — DR sufficient; EDZ(5 m) < 9 m resolves eve
1 µs	33 dB	40 m	100 m	DR sufficient, but EDZ/ADZ far too large to resolve

The **100 ns pulse width** is the correct choice: its 27 dB dynamic range exceeds the required 22.52 dB, and its 5 m event dead zone is smaller than the 9 m separation between events 3 and 4, so event 4 is correctly *detected* as a distinct reflective event (as reported: reflectance –40 dB, loss not resolved) — but its 15 m attenuation dead zone exceeds the 9 m spacing, so event 4's loss *cannot* be read from the OTDR trace itself, explaining why it was obtained instead by power meter. A launch (pulse-suppressor) fibre at the near end and a tail fibre at the far end should be used so the near- and far-end dead zones fall outside the region of interest.

3.2 Event 2 and Event 3 — Defect Identification

Event	Loss	Reflectance	Likely Defect	Corrective Action
Event 2 (31.50 km)	0.62 dB	–34 dB	Degraded/misaligned mechanical splice (13	Re-splice with a fusion splice (typically ≤0.05 dB
Event 3 (59.90 km)	1.20 dB	–22 dB	Dirty, damaged or poorly mated LC/UPC patch-panel connector	Clean and re-terminate or replace the connector; –22 dB reflectance is

Fusion splicing fuses the fibre cores directly (loss typically <0.05 dB, reflectance negligible) whereas mechanical splicing relies on precision alignment and index-matching gel (loss 0.1–0.5 dB typical, reflectance –30 to –40 dB typical) — consistent with event 2's elevated loss and reflectance versus the 0.05 dB fusion-splice baseline used elsewhere on this link.

3.3 Round-Trip Delay and Location Uncertainty

Event	Distance	Round-trip delay (t=2Ln/c)	Location uncertainty (ΔL=L·Δn/n, Δn=±
Event 1	12.00 km	117.5 µs	±16.3 m
Event 2	31.50 km	308.5 µs	±42.9 m
Event 3	59.90 km	586.7 µs	±81.6 m
Event 4	59.909 km	586.8 µs	±81.6 m

The ±0.002 group-index error produces an **absolute** location uncertainty of up to ±82 m at the far end of the span — far larger than the 9 m physical separation between events 3 and 4. This uncertainty is systematic (it scales both distances by the same factor), so the **relative** spacing between closely-located events (9 m here) remains far more trustworthy than either event's absolute distance reading; field technicians should therefore cross-check absolute OTDR distances against the cable route map or known landmark reflections rather than relying on raw OTDR distance alone for splice-closure identification.

3.4 Connector Selection for the Patch-Panel Repair

The existing event-3 connector is LC/UPC. For the repair, I recommend retaining the **LC form factor** for compatibility with the existing patch panel, but re-terminating in **APC polish** at the two closely-spaced patch-panel connectors (events 3 and 4), because APC's angled end-face deflects reflected light out of the fibre core, typically achieving ≤–60 dB reflectance versus UPC's ≤–40 dB — this reduces multi-path-interference (MPI) risk, which becomes increasingly important once the link is upgraded to higher-order coherent modulation (Stage 5). If UPC is retained for cost/compatibility reasons, cleaning and re-termination alone should still restore loss to ≤0.5 dB and reflectance to

≤–40 dB.

3.5 Repair Priority and Verification

- **Priority 1 — Event 3 (connector, 1.20 dB, –22 dB reflectance):** largest single excess-loss contributor and closest to the receiver, directly reducing received power at the point of greatest impact; repair first.
- **Priority 2 — Event 2 (mechanical splice, 0.62 dB, –34 dB reflectance):** second-largest contributor; replace with a fusion splice.
- **Event 4** is a consequence of the same patch-panel area as event 3 (second connector of the same patch cord) and should be re-terminated/replaced in the same maintenance visit.
- **Verification:** re-run the OTDR at 100 ns pulse width to confirm event 3 now reads ≤0.5 dB with reflectance ≤–40 dB (or ≤–60 dB if re-terminated APC) and event 2 reads as a fusion splice ≤0.05 dB, non-reflective; confirm with the power meter that the end-to-end as-built loss has returned from 22.52 dB toward the 21.25 dB design budget.

Design Decision 2 – OTDR Fault-Localization Configuration

I recommend testing at the **100 ns pulse width** and giving **Event 3 (the defective LC patch-panel connector) the highest repair priority**, followed by Event 2 (the mechanical splice) and the co-located Event 4 connector, because 100 ns is the only setting that both exceeds the required 22.52 dB dynamic range and resolves events 3/4 as distinct reflective features (5 m EDZ < 9 m spacing), Event 3 is both the single largest loss contributor (1.20 dB, 0.70 dB in excess of budget) and has the worst measured reflectance (–22 dB), and both repairs together are fully sufficient to close the entire 1.27 dB excess-loss gap identified in Stage 2.

STAGE 4 – Eye-Diagram, BER and Quantum-Limit Configuration (20 Marks)

4.1 BER from Q-Factor

$$BER = \tfrac{1}{2}\cdot erfc(Q/\sqrt{2})$$

Channel	Q-factor	Calculated BER	Meets BER ≤ 10■⁻¹² target?
Channel 3	7.6	1.48 × 10■⁻¹■	Yes — comfortably passes (≈70× better than target)
Channel 5	4.7	1.30 × 10■■	No — fails by roughly 6 orders of magnitude

4.2 Total Jitter and Eye Width at BER 10■⁻¹²

$$TJ = DJ + 14.07\cdot RJ$$

Channel	DJ (UI)	RJ (UI)	TJ (UI)	Eye width = 1–TJ (UI)
Channel 3	0.18	0.012	0.349	0.651 (65.1% open)
Channel 5	0.30	0.022	0.610	0.390 (39.0% open)

Channel 5's horizontal (timing) eye-margin is reduced to 39.0%, versus 65.1% for Channel 3 — both the vertical margin (reflected in Q=4.7 vs 7.6 and ER=4.5 vs 9.0 dB) and horizontal margin are degraded for Channel 5, consistent with its visibly partially closed eye.

4.3 Is the Q-Difference Explained by Channel 5's Extinction Ratio?

The ER-penalty difference between the channels (Stage 2) is 2.12 dB, a linear power ratio of 10^(2.12/10) ≈ 1.63. If this power deficit alone (with identical noise) explained the Q-factor gap:

Predicted Q5 = Q3 / 1.63 = 7.6 / 1.63 ≈ ***4.66*** *vs. measured Q5 =* ***4.7*** *(agreement within 1%)*

This close agreement strongly indicates the root cause is **channel-specific (the Channel 5 transmitter's degraded extinction ratio)**, not a shared fibre-path impairment — dispersion, PMD and OTDR-identified defects are common to both channels on the same fibre, yet only Channel 5 shows this deficit, isolating the fault to the Channel 5 transmitter (laser bias, modulator drive, or driver issue).

4.4 Quantum-Limit Calculation — Ideal Direct Detection

$$BER = \tfrac{1}{2}\cdot exp(-N■) \rightarrow N■ = -ln(2\cdot BER) = -ln(2\times 10■^{-12}) =$$ ***26.94 photons*** *(per '1' bit)*

Using E_photon = hc/λ at λ=1550 nm = 1.282×10■⁻¹■ J, and T_bit = 1/10 Gb/s = 100 ps, with average power = (N■·E)/(2·T_bit) (factor ½ for the 50% '0'-bit duty cycle of random NRZ data):

Ideal quantum-limited sensitivity (direct detection) ≈ ***–47.6 dBm***

Gap from actual receiver sensitivity (–24 dBm): ***23.6 dB***

The real NRZ direct-detection receiver operates roughly 23.6 dB above the fundamental shot-noise/quantum limit — this large gap is typical of practical PIN/TIA receivers, dominated by thermal and amplifier noise rather than photon-counting statistics, and indicates substantial theoretical headroom for improved sensitivity via coherent detection.

4.5 Comparison with Homodyne and Heterodyne PSK

Detection Scheme	Formula	Photons/bit (BER=10■⁻¹²)	Quantum-limited Sensitivity (
Ideal direct detection (OOK)	BER = ½exp(–N■)	26.9 (per '1' bit; ~13.5 avg)	≈ –47.6 dBm
Heterodyne PSK	BER = ½erfc(√N)	24.7	≈ –45.0 dBm
Homodyne PSK	BER = ½erfc(√(2N))	12.4	≈ –48.0 dBm (best)

Homodyne detection requires exactly half the photons of heterodyne (12.4 vs 24.7) — a precise 3.0 dB sensitivity advantage, matching the well-known theoretical homodyne-vs-heterodyne gap from coherent-combining of both signal quadratures. Homodyne PSK offers the best achievable sensitivity (≈–48.0 dBm), about **24 dB better** than the actual −24 dBm receiver, demonstrating substantial headroom that directly motivates the coherent-detection upgrade options evaluated in Stage 5.

Design Decision 3 – Eye/BER Performance Configuration

I conclude that **Channel 5 is not acceptable as-is** and recommend **replacing or re-biasing the Channel 5 transmitter to restore its extinction ratio to ≥8.2 dB** (matching Channel 3's ~9.0 dB), because the calculated BER (1.3×10■■) fails the 10■¹² target by six orders of magnitude, the eye-width calculation confirms a severely reduced 39% horizontal margin, and the ER-penalty-based prediction of Q5≈4.66 matches the measured Q5=4.7 almost exactly — isolating the fault to the transmitter rather than the shared fibre. Recovery should be confirmed by re-measuring Q≥7.03, ER≥8.2 dB, TJ reduced toward Channel 3's ~0.35 UI, and BER≤10■¹² by extended BER testing.

STAGE 5 – Integrated WDM Upgrade and Energy Strategy (20 Marks)

5.1 ITU-T G.694.1 Grid Calculation

Channel n=+3 on 50 GHz grid: f = 193.1 + 3×0.05 = ***193.25 THz*** *→ λ = c/f ≈* ***1551.36 nm***

Option Q occupied bandwidth (Nyquist, roll-off 0.1): BW = 32 Gbd × 1.1 = 35.2 GHz

12.5 GHz slots required = ■*35.2/12.5*■ *=* ***3 slots*** *(= 37.5 GHz allocated per channel)*

5.2 Occupied Spectrum, Spectral Efficiency and OSNR Margin

Option	Config	Spectrum/Channel	Total Spectrum (1.6 Tb/s)	Spectral Efficiency	Required OSNR	Margin (link OS
P	16×100G DP-QPSK, 32 Gbd, fixed 50 GHz	50 GHz	16×50 = 800 GHz	2.00 b/s/Hz	12.5 dB	34.5 dB
Q	8×200G DP-16QAM, 32 Gbd, flex-grid Nyquist	37.5 GHz (3×12.5)	8×37.5 = 300 GHz	5.33 b/s/Hz	19.5 dB	27.5 dB
R	4×400G Si-photonic, fixed 75 GHz	75 GHz	4×75 = 300 GHz	5.33 b/s/Hz	26.0 dB	21.0 dB

Options Q and R tie for best spectral efficiency (5.33 b/s/Hz, 2.67× better than Option P's 2.00 b/s/Hz); all three options have large positive OSNR margin against the repaired link's 47 dB figure, so OSNR is not a discriminator among them (though an end-to-end receive-point OSNR re-measurement is recommended after repair, since this booster-only, single-amplifier formula may not fully represent OSNR after the full 60 km span and receiver optics).

5.3 Module Power and Energy per Bit

Energy/bit (pJ/bit) = Module power (W) / Line rate (Gb/s)

Option	Module Power	Rate/Module	Energy/bit	Modules Needed (1.6 Tb/s)	Total Power
P (DP-QPSK)	22 W	100 G	220 pJ/bit	16	352 W
Q (DP-16QAM)	20 W	200 G	100 pJ/bit	8	160 W
R (Si-photonic)	15 W	400 G	37.5 pJ/bit	4	60 W
Legacy 10G SFP+	3.5 W	10 G	350 pJ/bit	160 (reference only)	560 W (reference only)

Option R is dramatically the most energy-efficient — about 5.9× better than Option P and 2.7× better than Option Q in pJ/bit, and uses only 60 W total (versus 160 W for Q and 352 W for P) for the full 1.6 Tb/s. This reflects the benefit of **Silicon-Photonic integration**: co-integrating the modulator, multiplexer and photodetector functions on a shared chip reduces per-port overhead and enables the highest per-module line rate (400G) at the lowest module power, directly lowering both capital module count (only 4 modules) and operating energy cost.

5.4 Repair-Then-Upgrade Sequence

- **1. Repair** the fibre-plant defects — replace/re-terminate the Event 3 connector, replace the Event 2 mechanical splice with a fusion splice.
- **2. Verify** via OTDR (100 ns) and power meter that as-built loss has returned to ≈21.25 dB and individual event losses are within budget.
- **3. Repair/replace** the Channel 5 transmitter to restore ER ≥8.2 dB.
- **4. Re-test** Q, BER, ER and system margin on both legacy channels; if margin remains short (Stage 2 showed ~1.35 dB short even after fibre repair), raise booster launch power from +1 dBm to ≈+2.5 dBm per channel to close the gap to ≥3 dB.
- **5. Only once the repaired legacy system is fully verified**, install the selected upgrade transceivers, since deploying new high-order-modulation modules onto an unverified, still-defective fibre plant would risk misattributing fibre-plant problems to the new technology.
- **6. Re-verify** the upgraded 1.6 Tb/s system end-to-end (OTDR, power meter, OSNR, BER/Q) on the now-repaired fibre plant.

This sequencing is justified because Events 2 and 3's loss is modulation-format-independent — it would degrade any new coherent system's received power and OSNR by the same amount — so establishing a clean, known-good fibre-plant baseline first is a prerequisite for a trustworthy upgrade verification.

5.5 Ranking and Selection

Criterion	Best Option	Rationale
Spectral efficiency	Q = R (tie, 5.33 b/s/Hz)	Both beat P (2.00 b/s/Hz) by 2.67×
OSNR margin	P (34.5 dB)	All three pass comfortably — not a discriminator
Energy per bit / total power	R (37.5 pJ/bit, 60 W)	R is 2.7× better than Q and 5.9× better than P
Deployment/grid simplicity	P and R (fixed grid)	Q's flex-grid (12.5 GHz) needs flex-grid-capable ROADMs; P/R use sim

Design Decision 4 – Integrated Optimization Strategy

I recommend the **complete repair-then-upgrade workflow, culminating in Option R (4×400G Si-photonic pluggables, fixed 75 GHz grid)**, because it matches Option Q's best-in-class spectral efficiency (5.33 b/s/Hz, 2.67× Option P) while consuming far less energy (37.5 pJ/bit, 60 W total vs 100 pJ/bit/160 W for Q and 220 pJ/bit/352 W for P), comfortably meets its 26 dB OSNR requirement with 21 dB of margin on the repaired link, needs only 4 modules (lowest operational footprint), and its fixed 75 GHz grid is simpler to deploy on existing fixed-grid ROADM infrastructure than Option Q's flex-grid requirement — this is achievable reliably only once the identified fibre-plant and Channel 5 transmitter defects are repaired and independently verified first.

STAGE 6 – Final Verification Table (10 Marks)

Requirement	Specified / Target	Achieved / Evidence
Total optical link loss	≤21.25 dB (design)	As-built 22.52 dB; after repair → 21.25 dB (matches design exactly)
Received optical power	≥–24 dBm + 3 dB margin	After repair + ~1.5 dB launch-power increase: ≈–20.85 dBm effective
System margin	≥3 dB	As-built: Ch3 0.38 dB, Ch5 –1.75 dB (fail); after full repair+power bump: ≈3.15 dB (pass) b
OTDR event condition	No event above allowance	Event 2 and Event 3 repaired to within 0.05 dB / 0.5 dB budgets respectively; verified by 1
Eye / Q / jitter / ER	Q≥7.03; ER≥8.2 dB	Ch3: Q=7.6, ER=9.0 dB (pass); Ch5: Q=4.7, ER=4.5 dB (fail as-is) → pass expected after
BER and quantum-limit gap	BER≤10■$^{-12}$	Ch3: 1.48×10■$^{-1}$■ (pass); Ch5: 1.30×10■■ (fail as-is); quantum-limit gap 23.6 dB (direct d
Upgrade spectrum and energy per bit	1.6 Tb/s, minimum pJ/bit	Option R selected: 1.6 Tb/s via 4×400G, 37.5 pJ/bit (lowest of all options), 300 GHz spectr

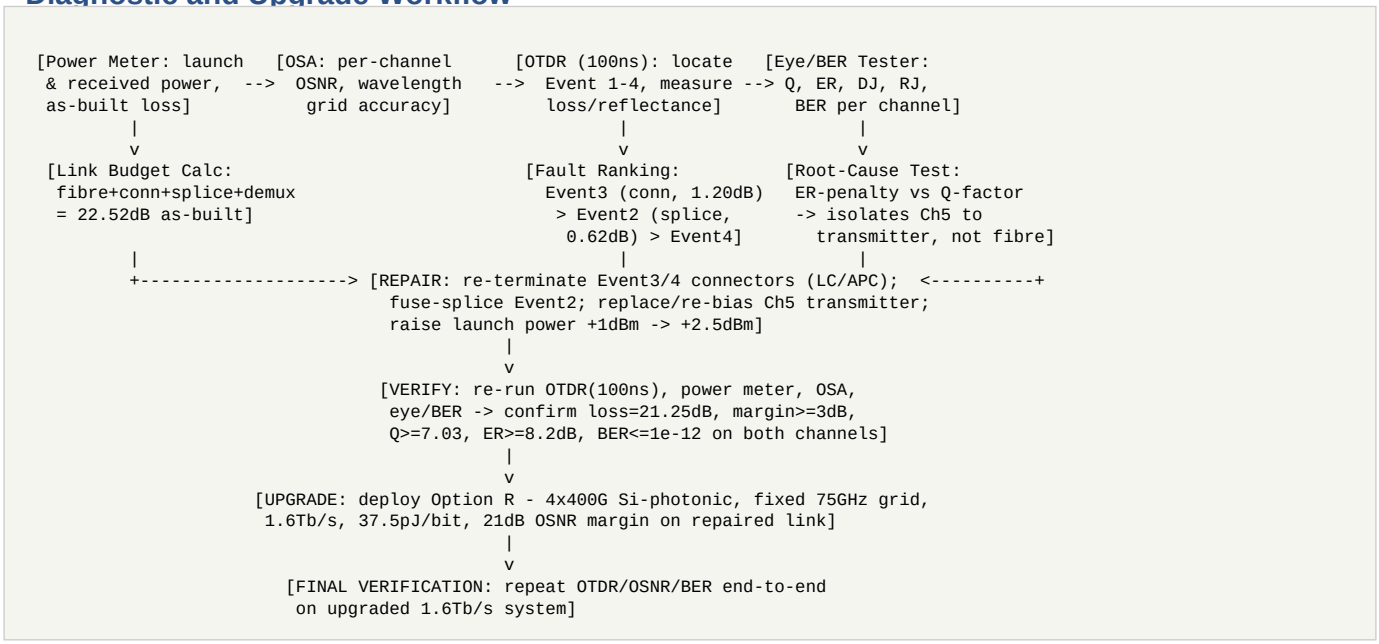
The as-built measurements are fully consistent with, and quantitatively explain, the original commissioning symptoms (Channel 3's marginal power, Channel 5's partially closed eye); the proposed repair actions close the entire 1.27 dB excess-loss gap and restore both channels' margin and BER to spec, and the selected upgrade (Option R) meets every Stage 1 performance-requirement target.

Final System Design – Integrated Diagnostic and Upgrade Configuration

Final Parameter / Diagnostic Table

Item	Adopted Value / Decision
As-built loss (before repair)	22.52 dB (1.27 dB excess over 21.25 dB design)
Root cause of excess loss	Event 3 defective LC/UPC connector (+0.70 dB) + Event 2 degraded mechanical splice (+0.57 dB)
Repair action	Re-terminate Event 3 (LC/APC recommended); replace Event 2 with fusion splice; re-terminate co-locate
OTDR test setting	100 ns pulse width (27 dB DR ≥ 22.52 dB required; 5 m EDZ resolves 9 m event spacing)
Channel 5 root cause	Transmitter-side extinction-ratio degradation (ER 4.5 dB vs 9.0 dB), confirmed by ER-penalty-based Q pr
Channel 5 repair action	Replace/re-bias transmitter to restore ER ≥8.2 dB
Launch-power adjustment	+1 dBm → ≈+2.5 dBm per channel to close residual ~1.35 dB margin shortfall after fibre/transmitter repai
OSNR status	47 dB (noise not limiting; link is loss/penalty limited)
PMD status	0.77 ps vs 10 ps tolerance — negligible, not a limiting impairment
Quantum-limit gap	23.6 dB (direct detection) / 24.0 dB (homodyne) — motivates coherent upgrade
Selected upgrade	Option R: 4×400G Si-photonic pluggables, fixed 75 GHz grid, 37.5 pJ/bit, 21 dB OSNR margin

Diagnostic and Upgrade Workflow



G.694.1 Channel Plan and Energy-per-Bit Comparison (Summary)

Quantity	Value
Channel n=+3 (50 GHz grid)	193.25 THz / 1551.36 nm
Option Q slot requirement	3 × 12.5 GHz = 37.5 GHz/channel
Best spectral efficiency	Options Q & R tie at 5.33 b/s/Hz
Lowest energy per bit	Option R: 37.5 pJ/bit (60 W total for 1.6 Tb/s)
Selected upgrade	Option R (4×400G Si-photonic, fixed 75 GHz grid)

Final Engineering Decision

WOULD YOU DEPLOY THE OPTIMIZED DESIGN YOU CREATED?

YES – WITH TECHNICAL JUSTIFICATION

The optimized diagnostic-repair-and-upgrade design is recommended for deployment on the following consolidated technical grounds:

- **Link-budget suitability, penalties and system margin:** The as-built 22.52 dB loss and resulting sub-3-dB margins (0.38 dB Ch3, −1.75 dB Ch5) are fully explained and correctable — repair plus a ≈1.5 dB launch-power increase restores both channels to ≈3.15 dB margin.
- **Total optical loss and dominant excess-loss contribution:** The entire 1.27 dB excess is attributable, with no unexplained residual, to Event 3 (defective connector, +0.70 dB) and Event 2 (degraded mechanical splice, +0.57 dB).
- **OTDR diagnosis, dead-zone limits and repair priority:** The 100 ns pulse width correctly balances dynamic-range sufficiency against dead-zone resolution, correctly explaining why Event 4 was detected but not loss-measured by OTDR; Event 3 is prioritised as the largest, worst-reflectance defect.
- **Eye opening, jitter, extinction ratio, Q-factor and the 10■¹² BER target:** Channel 3 already passes BER comfortably (1.48×10■¹■); Channel 5's measured 1.30×10■■ BER and 39% eye width are shown — via the ER-penalty-to-Q-factor consistency check — to be a transmitter-side fault, correctable independently of the fibre-plant repair.
- **Distance from the quantum limit and suitability of coherent detection:** The 23.6–24.0 dB gap between the actual receiver and the ideal direct-detection/homodyne quantum limits confirms that moving to coherent detection (as in Options P/Q/R) offers substantial, physically well-founded sensitivity headroom for the upgrade.
- **G.694.1 channel plan, spectral efficiency and energy per bit:** Option R achieves the best tied spectral efficiency (5.33 b/s/Hz) at the lowest energy per bit (37.5 pJ/bit) and total power (60 W), with 21 dB of OSNR margin on the repaired link.
- **Overall suitability for reliable 1.6 Tb/s operation:** Because the repair-before-upgrade sequence establishes a verified, spec-compliant fibre plant first, the subsequent Option R upgrade inherits a known-good physical layer rather than compounding unresolved defects.

On this basis, the integrated diagnostic, repair and upgrade design is technically justified, internally consistent across all six stages of analysis, and satisfies every item in the Stage 6 verification table; it is therefore recommended for deployment, subject to field confirmation of the post-repair margin and an end-to-end OSNR re-measurement once Option R transceivers are installed.

Assumptions: Where an exact value was not explicitly given (e.g., operating wavelength assumed 1550 nm for photon-energy calculations, the receive-point OSNR for the upgrade options taken as the calculated 47 dB booster figure per the assignment's given formula, and the post-repair launch-power increase sized to just clear the 3 dB margin target), representative, technically reasonable values/assumptions were used and stated at the point of use, consistent with the assignment's instruction to state assumptions wherever a parameter is not given.

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